EXAMINATION NUMBER: _____

MARIST SECONDARY SCHOOL
P.O. Box 46
Malirana Tel: (+265) 0994484899/0991587988/0887251944
Dedza, MALAWI E-mail: maristhead@gmail.com

MARIST SECONDARY SCHOOL

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS

PHYSICS

Subject Number: M164/II

Time Allowed: 2 hour sessions

Thursday, 25th May

Paper II

(40 marks)

Practical

Instructions

1. This paper contains **8** printed pages. Please check.
2. Fill your examination number at the top of each page.
3. Answer all the **4** questions in the spaces provided.
4. Use of electronic calculators is allowed.
5. The maximum number of marks for each answer is indicated against each question.

EXAMINATION NUMBER: _____

1. a. With the aid of a well labelled diagram, describe an experiment that could be done to show that liquids expand by different amounts when heated equally.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

7 marks

EXAMINATION NUMBER: _____

2. a. With the aid of a diagram describe an experiment that can be carried out to show that liquid pressure increases with depth.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

7 marks

EXAMINATION NUMBER: _____

b. With the aid of a labelled diagram describe an experiment that can be done to verify the principle of moment of forces.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

6 marks

EXAMINATION NUMBER: _____

3. You are provided with a clamp and clamp stand, stop watch, a 100cm string, a 100g mass and a 1metre rule.

a. Tie the 100g mass to one end of the string.

Excel Bk3 page 324 F14.4

b. Arrange the apparatus as shown in the figure below.

c. Pull the mass sideways by a small angle and release it to oscillate. Start the stop watch immediately.

d. Record the time taken for 10 oscillations.

e. Calculate the periodic time and frequency.

f. Repeat procedure a to e with length of string changed to 35cm, 40cm, 45cm and 50cm.

Length of string(cm)	**Time taken for 10 oscillations(seconds)**	**Periodic time (T), (seconds)**	**Frequency (f), (Hertz)**
30			
35			
40			
45			
50			

6 marks

g. Plot a graph of frequency against mass

(graph full page)

EXAMINATION NUMBER: _____

4. You are provided with a matchbox containing three unknown materials R, P and Q with terminals protruding out, 3 connecting wires, an ammeter, two cells and a cell holder.

a. Make sure that the match box is looking as shown below. Do not open the match box.

Marist2022 Mock2 Page 3

b. Connect the terminals of unknown material marked R on the gap XY and observe the ammeter reading. Record the results in the table below by indicating '***reading'*** or '***no reading'***.

c. Reverse the connection of the same material on the gap and observe the ammeter reading. Record the results in the table.

d. Repeat steps b and c with unknown components P and Q.

Unknown material	Ammeter reading in first connection	Ammeter reading when reversed
R		
P		
Q		

3 marks

d. Identify the materials as either a diode, insulator or a resistor

R : _____

P : _____

Q : _____

3 marks

e. Give reasons for each of your choices in ***d*** above

_____

_____

_____

EXAMINATION NUMBER: _____

_____

_____

3 marks

f. Name an electronic component used as to store charge.

_____

1 mark

END OF QUESTION PAPER